

HUMAN-COMPUTER INTERACTION (HCI)

FINAL TERM EXAM COMPREHENSIVE STUDY GUIDE & PRACTICE

Syllabus Scope: Weeks 6, 7, 8, 9, 10 & 11 | Midterm & Final Pattern Aligned

University of Central Punjab (UCP) — Faculty of Information Technology

Course Code & Title:	Human-Computer Interaction (HCI)	Target Exam:	Final Term Exam (40% Weightage)
Instructor:	Dr. Noman Nazar / Dr. Nadeem Ahmad	Question Types:	MCQs, Short-Answer, Scenario-Based
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EXAM PREPARATION STRATEGY: This comprehensive study guide is specifically crafted for the Final Term Examination in Human-Computer Interaction at UCP. It synthesizes all lecture slide sets from **Week 6 (Prototyping & HCD)**, **Week 7 (Efficiency & KLM/Fitts's)**, **Week 8 (Learnability & Design Principles)**, **Week 9 (Perceptual Organization & Gestalt)**, **Week 10 (Evaluation Methods & Heuristics)**, and **Week 11 (Cognitive Walkthrough)**. Review the theoretical foundations, formulas, guidelines, and solved scenario questions to maximize your performance.

1. Prototyping: Lo-Fi vs. Hi-Fi & Human-Centered Design

In Interaction Design, a **prototype** is a limited, concrete representation of a product design built to test concepts, clarify vague requirements, and gather feedback from stakeholders. Prototypes answer design questions and help choose between alternatives.

Prototyping Techniques & Types

- **Throwaway / Rapid Prototyping:** Built purely to explore ideas and gather initial feedback. Once evaluated, it is discarded and does not form part of the final codebase.
- **Evolutionary Prototyping (RAD):** An initial prototype is presented to users, evaluated, and incrementally refined in cycles. Each stage evolves directly toward the final production system.
- **Low-Fidelity (Lo-Fi) Prototyping:** Quick, cheap, easily changed representations (e.g., paper sketches, storyboards, 8-up folds, card sorting, Wizard of Oz).
- **High-Fidelity (Hi-Fi) Prototyping:** Usually software-based; looks, feels, and operates much like the final system with active mouse/keyboard interactions.

Comparison: Low-Fidelity vs. High-Fidelity Prototypes

Prototyping Type	Key Advantages	Key Disadvantages & Risks
Low-Fidelity (Lo-Fi) (<i>Paper sketches, Storyboards</i>)	- Lower development cost & fast - Evaluates multiple design concepts - Users focus on big picture (structure/function) - Easily disposable; developers don't get over-attached	- Limited error checking & interaction - Poor detailed specification for coding - Facilitator-driven during user tests - Limited utility after requirements set
High-Fidelity (Hi-Fi) (<i>Figma, Interactive Code</i>)	- Complete functionality & interactive - User-driven exploration - Looks & feels like final product - Living specification & sales/marketing tool	- Expensive & time-consuming to create - Inefficient for initial requirements gathering - Users focus on small details (font size/color) - Developers resist changes due to time invested

EXAM TIP: LO-FI ADVANTAGE: Why Lo-Fi is better early in design: Hi-Fi prototypes make users comment on superficial details (like button colors) rather than structure, and make developers over-attached to their code. Always get Lo-Fi right before investing in Hi-Fi!

2. Learnability & The Human Information Processing System

How People Learn Interfaces & The Abowd & Beale Framework

Users learn primarily by **doing**—they have a specific goal to achieve (e.g., 'remove red-eye from photo') and explore the interface to satisfy that goal. Their primary goal is **not** learning the software itself. Interaction is modeled by the **Abowd & Beale Framework** across four components: **System (S)**, **User (U)**, **Input (I)**, and **Output (O)**. The interaction cycle moves through *Articulation* (User specifies task at Input), *Performance* (Input translates to System core), *Presentation* (System outputs state), and *Observation* (User interprets Output).

Model Human Processor (MHP) & Cognitive Architecture

The Model Human Processor (Card, Moran & Newell) models human cognition via three interacting processors, each with a characteristic cycle time (\$T\$):

Processor Type	Cycle Time (\$T\$)	Function & Exam Significance
Perceptual Processor	$T_p \approx 100\text{ ms}$ (Range: 50–200 ms)	Organizes physical stimuli from sensory organs and creates a scene. Perceptual Fusion: Two stimuli occurring within $T_p \approx 100\text{ ms}$ are perceived as a single fused event (e.g., frame rate in movies creating continuous motion).
Cognitive Processor	$T_c \approx 70\text{ ms}$ (Range: 30–100 ms)	Compares stimuli, retrieves memories, and selects responses. Operates via Rule-based (Novices, IF-THEN rules), Knowledge-based (Unfamiliar problems), or Skill-based (Expert automatic behavior).
Motor Processor	$T_m \approx 70\text{ ms}$ (Range: 25–170 ms)	Drives physical muscle movements. Runs in Open Loop Control (no feedback, $T_m \approx 70\text{ ms}$) or Closed Loop Control (feedback-dependent, $T_p + T_c + T_m \approx 240\text{ ms}$).

Human Memory Hierarchy & Chunking

- **Sensory Memory:** Buffer for incoming stimuli (200–500 ms). Reflexes & filtration occur here pre-attentively.
- **Short-Term / Working Memory (STM):** Limited capacity of **\$7 \pm 2\$ chunks** (or \$5 \pm 2\$ in complex UI tasks). Rapid decay of **~10 seconds**. Fended off by *maintenance rehearsal* (which costs attention).
- **Long-Term Memory (LTM):** Infinite capacity. Information is transferred from STM via *elaborative rehearsal* (building connections to existing chunks).
- **Chunking & Connections:** A chunk is a meaningful unit of memory. Information presentation and **Familiarity** dictate chunking efficiency (e.g., 'BMW RCA AOL' is recalled instantly as 3 chunks, whereas 'M W B C R A' requires 7+ separate slots).

Recognition vs. Recall

Recognition (Knowledge in the World): Remembering with the assistance of a visible cue (e.g., selecting a command from a dropdown menu). Recognition is far easier for humans.

Recall (Knowledge in the Head): Remembering information with zero outside cues (e.g., typing a complex command in Unix CSH). High memory load.

3. Fundamental Design Principles

Norman's 6 Fundamental Design Principles

Principle	Definition & Mechanics	UI Application / Example
Affordance	Perceived and actual properties of an object that visually signal how it can be used. Learned from experience.	3D raised buttons afford pushing; sliders afford dragging; door handles afford pulling.
Mapping	The spatial/conceptual relationship between controls and their actual real-world effects on the system.	Stove burner knobs arranged in the exact 2x2 layout of the physical burners.
Feedback	Immediate, visible/audible/haptic indication of the system's reaction to user input. Bridges Gulf of Evaluation.	Button depressing when clicked, spinner on page load. Response Times: <0.1s (instant), 1s (uninterrupted flow), 10s (attention limit).
Visibility	Relevant parts and current state of the system should be clearly visible to the user.	Breadcrumb navigation paths (<i>Home > News > World</i>) showing current hierarchy.
Constraints	Restricting the range of possible user actions to prevent errors before they happen.	Disabling past dates in a calendar widget or using dropdowns instead of free text entry.
Consistency	Principle of Least Surprise: Similar things should look and act the same; different things should look different.	Internal: Same button layout across all windows. External: Following platform standards (Ctrl+S for save).

4. Efficiency: Fitts's Law, Steering Law & KLM-GOMS

Fitts's Law (Pointing Model)

Fitts's Law is a mathematical model of fine motor control that predicts the **Movement Time (\$MT\$)** required to rapidly point to a target area as a function of the target's distance (**Amplitude - \$A\$**) and width (**Size - \$W\$**):

Index of Difficulty (ID): $\text{ID} = \log_2\left(\frac{2A}{W}\right) + c$ (where $c = 0, 0.5, \$$ or $\$1$)

Movement Time (MT): $\text{MT} = a + b \times \text{ID}$ (where a is reaction time, b is device constant)

Key Implications of Fitts's Law:

- **Bigger & Closer is Faster:** Increasing target width (W) or decreasing distance (A) reduces MT .
- **Screen Edges & Corners (Infinite Width):** Screen edges/corners stop the mouse cursor from overshooting. Since $W \rightarrow \infty$, $\text{ID} \rightarrow 0$, making edges/corners the fastest targets on a screen.
- **Macintosh vs. Windows Menu Bar:** Mac menu bar sits at the top edge of the screen ($W = \infty$, fast acquisition). Windows menu bars float inside app windows (fixed W , slower acquisition).
- **Mile-High Menus & Magic Corners:** Placing critical controls (like Start button) in screen corners maximizes pointing efficiency.
- **Functional Aggregation:** Grouping frequently used controls close together minimizes movement distance A .

Steering Law (Tunneling & Cascading Menus)

Steering involves constraining mouse movement through a tunnel of length D and width S (e.g., navigating cascading submenus):

Steering Formula: $T = a + b \left(\frac{D}{S}\right)$

CRITICAL DIFFERENCE: Steering difficulty is **linear** (D/S), whereas Fitts's Law pointing is **logarithmic** ($\log_2$).

Because difficulty grows linearly, steering tasks are dramatically slower and error-prone. This is why cascading submenus are considered bad UI design.

Keystroke Level Model (KLM / KLM-GOMS)

KLM predicts expert execution time for routine, error-free tasks by summing physical and mental operator execution times:

Operator	Description	Standard Time (\$s\$)	Rules & Exam Notes
K	Keystroke / Key Press	0.28 s (Avg) 0.08s (Best) / 1.2s (Worst)	Standard typist = 0.28s. Pressing single letter or enter.
B	Mouse Button Press/Release	0.1 s	Full Mouse Click (Down + Up) = 2B = 0.2s .
P	Pointing Mouse to Target	1.1 s	Moving cursor to target via Fitts's Law.
H	Homing Hands	0.4 s	Switching hand between keyboard and mouse.
D	Drawing Lines	1.1 s	Drawing n domain line segments.
M	Mental Preparation	1.2 s	Cognitive pause before action. Applied when initiating task, making strategy decision, retrieving chunk, or verifying.

5. Design Guidelines & Standards

Shneiderman's 8 Golden Rules of Interface Design

- 1. **Strive for consistency:** Consistent sequences of actions, terminology, layout, and fonts.

- 2. **Enable frequent users to use shortcuts:** Provide macros, function keys, and hidden accelerators for experts.
- 3. **Offer informative feedback:** For every user action, the system should respond appropriately.
- 4. **Design dialogs to yield closure:** Organize sequences of actions with a beginning, middle, and end (e.g., checkout confirmation).
- 5. **Offer error prevention and simple error handling:** Design so users cannot make serious errors; offer clear recovery steps.
- 6. **Permit easy reversal of actions:** Support Universal Undo to relieve user anxiety.
- 7. **Support internal locus of control:** Make users feel they are in control of the system, not trapped by it.
- 8. **Reduce short-term memory load:** Keep displays simple; promote recognition over recall.

Norman's 7 Principles of Design

- 1. **Use both knowledge in the world and knowledge in the head:** Combine visible UI cues with internal mental models.
- 2. **Simplify the structure of tasks:** Reduce unnecessary cognitive complexity.
- 3. **Make things visible:** Bridge the **Gulf of Execution** (how to do it) and **Gulf of Evaluation** (what state it's in).
- 4. **Get the mappings right:** Ensure natural spatial and physical relationships between controls and outcomes.
- 5. **Exploit the power of constraints:** Use physical, logical, semantic, and cultural constraints.
- 6. **Design for error:** Assume errors will happen; make them recoverable and non-catastrophic.
- 7. **When all else fails, standardize:** If natural mappings don't exist, create universal standards.

6. Perceptual Organization & Gestalt Laws

Pre-Attentive Processing vs. Feature Conjunction Search

Pre-Attentive Processing (<200–250 ms): Visual properties processed rapidly and in parallel by the low-level visual system without conscious focus. Properties include line orientation, length, size, curvature, color/hue, density, contrast, and 3D depth cues.

Conjunction of Features: When a target shares multiple features with distractors (e.g., searching for a *red circle* among *red squares* and *blue circles*), pre-attentive pop-out fails. The user must perform a slow, **sequential visual search**.

Gestalt Principles of Visual Grouping

Gestalt Law	Perceptual Rule	UI Design Application
Proximity	Objects close to each other are perceived as a group.	Place related form labels close to their input boxes; separate form sections with white space.
Similarity	Objects sharing visual characteristics (shape, color, size) appear grouped.	All primary buttons styled with the same green background; secondary buttons styled in grey.
Continuity	The eye follows smooth continuous lines/paths rather than abrupt breaks.	Aligning form fields and table columns along vertical grid lines to guide the scanning eye.
Closure	The mind automatically fills in missing visual gaps to perceive a whole shape.	Effective use of 'White Space' and icon outlines without drawing complete bounding boxes.
Figure-Ground	The visual field is separated into focal figures (foreground) and background.	Modal dialog popups dimming/blurring the background screen to focus total attention.
Common Fate	Elements moving together in the same direction/velocity are perceived as grouped.	Selecting multiple icons and dragging them; semi-transparent copies move together.
Connectedness	Elements physically connected by lines/borders are perceived as more related than unlinked items.	Flowcharts or wizard step connectors linking numbered progress nodes.
Prägnanz (Simplicity)	Complex stimuli are reduced to the simplest possible symmetrical shapes.	Minimalist grid card layouts on dashboards to reduce cognitive visual clutter.

7. Evaluation Methods & Its Classification

Classification by Evaluation Timeline & Purpose

Evaluation Type	Timing in Development	Primary Purpose & Data Collected
Exploratory	Before Design	Explores user needs and design space. Collects observational & usage data (e.g., diary studies, ethnography).
Predictive	During Early Design (No Prototype Required)	Uses engineering models (GOMS/KLM) to predict efficiency and task time without testing live users.
Formative	During Interface Development	Informs design and identifies <i>why</i> something went wrong. Collects qualitative process data .
Summative	After Interface Completion	Assesses overall quality against benchmarks. Collects quantitative bottom-line data (time, errors, success rate).

Usability Inspection vs. Usability Testing

Usability Inspection (Expert-Based / Discounted Usability): Usability specialists inspect the interface guided by heuristics or task scenarios without involving real end-users (e.g., Heuristic Evaluation, Cognitive Walkthrough). Cheap and fast.

Usability Testing (User-Based): First-hand empirical testing of interface designs with representative end-users performing real tasks (e.g., thinking aloud, controlled lab experiments).

8. Nielsen's 10 Usability Heuristics

- 1. Visibility of System Status (Provide Feedback):** Keep users informed about system state within reasonable time. Response times: <0.1s (instant), 1s (uninterrupted flow), 10s (attention limit; show progress bar).
- 2. Match Between System and Real World:** Speak the user's language with familiar words, phrases, and concepts rather than system jargon (e.g., 'Cannot open chapter 5' vs 'Error 0x80070002').
- 3. User Control and Freedom (Clearly Marked Exits):** Provide emergency exits (Cancel, Undo, Redo, Quit) so users can leave unwanted states without feeling trapped.
- 4. Consistency and Standards:** Follow platform conventions. Words, actions, and UI elements should mean the same thing throughout the system (Principle of Least Surprise).
- 5. Recognition Rather Than Recall:** Minimize short-term memory load by making objects, actions, and options visible. Provide defaults, examples, and visible instructions.
- 6. Flexibility and Efficiency of Use (Accelerators):** Provide shortcuts, macros, keyboard accelerators, and split menus for expert users while keeping the interface friendly for novices.
- 7. Aesthetic and Minimalist Design:** Remove rarely needed or irrelevant information. Extra clutter competes with relevant data and diminishes visibility.
- 8. Error Prevention:** Careful design that prevents errors from occurring in the first place (e.g., date-picker calendar widgets, reasonableness checks).
- 9. Help Users Recognize, Diagnose & Recover from Errors:** Error messages expressed in plain language (no codes), precisely indicating the problem and constructively suggesting a solution without blaming the user.
- 10. Help and Documentation:** Easy to search, task-focused, step-by-step help documentation (e.g., Tooltips, Wizards, Context-sensitive help).

Heuristic Evaluation Process & Severity Ratings

Process: 1. *Briefing Session* (Project manager briefs evaluators on goals, domain, personas). 2. *Evaluation Period* (Evaluators perform 2 passes independently: 1st pass for flow, 2nd pass against heuristic checklist). 3. *Debriefing Session* (Aggregate findings and compare).

Evaluator Curve: A single evaluator finds only ~35% of usability problems. Using **3 to 5 evaluators** catches ~75–80% of usability issues satisfactorily.

Severity Ratings (0 to 4 Scale):

- 0:** Not a usability problem at all.
- 1 (Cosmetic):** Fix only if extra time is available.
- 2 (Minor):** Low priority usability fix.
- 3 (Major):** Important to fix; high priority.
- 4 (Usability Catastrophe):** Imperative to fix before product release.

9. Evaluation Method: Cognitive Walkthrough

Cognitive Walkthrough is a formalized usability inspection method that evaluates an interface for **exploratory learning** (how easy it is for first-time users to perform tasks without training). An expert steps through the task sequence imitating a user's thought process.

4 Prerequisites Required Before Walkthrough:

- 1. Detailed description or prototype of the interface (exact menu wording matter!).
- 2. Description of target users and their prior background/experience.

- 3. Task description for a representative task.
- 4. Complete written action list required to complete the task.

The 4 Core Cognitive Walkthrough Questions (Asked at EVERY Action Step):

Q1: Will users see that the action is available? *(Is the control visible on screen?)*

Q2: Once found, will users know that it is the correct action needed? *(Is the label/icon recognizable and matched to their intent?)*

Q3: Is the effect of the action the same as the user's goal at that point? *(Does it align with their mental subgoal?)*

Q4: After the action, will users understand the feedback they get? *(Does system response confirm success or failure?)*

10. Final Term Exam Practice Questions & Scenario Solutions

Part A: High-Yield Multiple Choice Questions (MCQs)

1. In standard Keystroke Level Model (KLM) calculations, what is the default execution time allocated for a Mental operator (\$M\$)?

A) 0.28 seconds | B) 0.40 seconds | C) 1.10 seconds | D) 1.20 seconds

Correct Answer: D (1.20 seconds). The mental preparation operator \$M\$ is standardized at 1.2 s across KLM calculations.

2. Why does the Macintosh top menu bar offer faster pointing access than a floating Windows application menu bar according to Fitts's Law?

A) Mac menus are placed closer to the center | B) The screen top edge provides an effectively infinite target width ($W \rightarrow \infty$) | C) Mac uses open loop motor control | D) Windows menus violate Gestalt Continuity

Correct Answer: B. Screen edges prevent cursor overshooting, making target width effectively infinite and reducing \$MT\$.

3. Why are cascading submenus frequently cited as poor interface design in efficiency lectures?

A) They violate Fitts's Law | B) They require steering, where difficulty increases linearly (D/S) rather than logarithmically | C) They require 5 Mental operators | D) They lack affordances

Correct Answer: B. Steering difficulty is linear $ST = a + b(D/S)$, making tunneling through cascading submenus slow and error-prone.

4. Which heuristic evaluation severity rating is assigned to a problem that is imperative to fix before product release?

A) Severity 1 | B) Severity 2 | C) Severity 3 | D) Severity 4

Correct Answer: D (Usability Catastrophe). Severity 4 represents catastrophic usability blocks.

5. What is the typical cycle time (\$T_c\$) of the Cognitive Processor in Card et al.'s Model Human Processor?

A) ~100 ms | B) ~70 ms | C) ~240 ms | D) ~500 ms

Correct Answer: B (~70 ms). Perceptual processor $T_p \approx 100 \text{ ms}$, Cognitive $T_c \approx 70 \text{ ms}$, Motor $T_m \approx 70 \text{ ms}$.

Part B: Exam Short-Answer Questions & Model Answers

Q1: Differentiate between Formative and Summative Usability Evaluations.

Answer: Formative evaluation takes place *during* interface development to inform the design process and identify *why* issues occur; it collects **qualitative process data**. Summative evaluation takes place *after* interface completion to measure overall quality against benchmarks; it collects **quantitative bottom-line data** (e.g., task completion time, error count).

Q2: Explain the difference between Open Loop and Closed Loop Motor Control in human information processing.

Answer: Open Loop Control operates without sensory feedback ($T_m \approx 70 \text{ ms}$), running automatically for well-learned rapid muscle movements. Closed Loop Control relies on continuous perceptual feedback and decision adjustments ($T_p + T_c + T_m \approx 240 \text{ ms}$), used when precision or visual correction is required.

Q3: Why are 3 to 5 evaluators recommended for Heuristic Evaluation rather than just 1?

Answer: A single evaluator detects only ~35% of usability problems. Using 3 to 5 independent evaluators catches ~75–80% of total usability problems satisfactorily, balancing cost-effectiveness with high error coverage.

Part C: Final Exam Scenario-Based Question & Complete Solution

Exam Scenario Question: A bank's mobile app requires users to transfer money. When an error occurs during transfer, a modal popup displays: *'Error 0x800F0922: System exception on remote Apache socket.'* with an 'OK' button that closes the app. Analyze this design by identifying **three Nielsen Heuristics violated** and propose an improved redesign solution.

1. Heuristic Violated — Help users recognize, diagnose & recover from errors (Heuristic #9): The popup displays raw technical server codes ('Error 0x800F0922') rather than plain language. It offers no constructive recovery solution.

2. Heuristic Violated — Match between system & the real world (Heuristic #2): Uses developer jargon ('remote Apache socket') rather than natural user task terms.

3. Heuristic Violated — User control & freedom (Heuristic #3): Clicking 'OK' abruptly terminates the app without providing an emergency exit or retry option.

Redesign Solution: Display plain text: *'Money Transfer Unsuccessful. We couldn't connect to the bank network. Your account balance has NOT been deducted.'* Provide two clear action buttons: **[Retry Transfer]** (Primary) and **[Cancel & Go to Home]** (Secondary).